

Workshop Activities

Activity 1 – Crosstabulation with PivotChart and Slicers

This activity requires you to build a PivotChart for total sales with slicers for distribution centre, year ordered, and new customers, for the *EJB* data.

Open the *EJB* data file and complete the following steps:

- Step 1.
 - Select any cell in the range of the data
- Step 2.
 - Click the Insert tab on the ribbon
 - In the Charts group, click the PivotChart button
- Step 3.
 - When the Create PivotChart dialog box appears:
 - Under Choose the data that you want to analyze, choose Select a table of range and enter Table1 in the Table/Range: box
 - Under Choose where you want the PivotChart to be placed, select New Worksheet
 - Click OK
- Step 4.
 - Select any cell in the empty PivotTable
- Step 5.
 - In the PivotTable Fields task, under Choose fields to add to report:
 - Drag the \$ Sales field to the Values area
 - Drag the Distribution Centre field to the Rows area
 - Drag the Year Ordered field to the Rows area
 - Drag the New Customer? field to the Columns area
- Step 6.
 - Click the PivotChart, right-click and select Change Chart Type from the menu
 - In the Change Chart Type dialog box, select Column from the list of charts and then select Stacked Column from the gallery.
 - Click OK

The next step adds an Excel feature called Slicers.

- Step 7.

- With the PivotChart selected, in the PivotChart Analyse tab in the Ribbon, select Slicers in the Filters group
- In the Insert Slicers dialog box, select Distribution Centre, New Customer? and Year Ordered

The next steps format the Slicers and PivotChart.

- Step 8.
 - With the Slicer box for Distribution Centre selected, in the Slicer tab in the Ribbon:
 - In the Buttons group, enter 7 in the Columns: box
- Step 9.
 - With the Slicer box for Years Ordered selected, in the Slicer tab in the Ribbon:
 - In the Buttons group, enter 3 in the Columns: box
- Step 10.
 - With the Slicer box for New Customer? selected, in the Slicer tab in the Ribbon:
 - In the Buttons group, enter 2 in the Columns: box
- Step 11.
 - With the PivotChart selected, in the PivotChart Analyse tab in the Ribbon:
 - In the Show/Hide group, click the down arrow next to Field Buttons
 - Select Hide All
- Step 12.
 - Select the vertical axis labels, right-click and select Format Axis, and in the Format Axis task pane:
 - Select Axis Options and in the drop-down menu next to Display units:
 - Select Thousands
 - Deselect the check box next to Show display units label on chart

After positioning the Slicer boxes adjacent to the PivotChart and sizing all boxes to create a seamless display, a single Excel object can be created with the following step.

- Step 13.
 - Holding the Ctrl key, select the three Slicer boxes and PivotChart Right-click and select Group

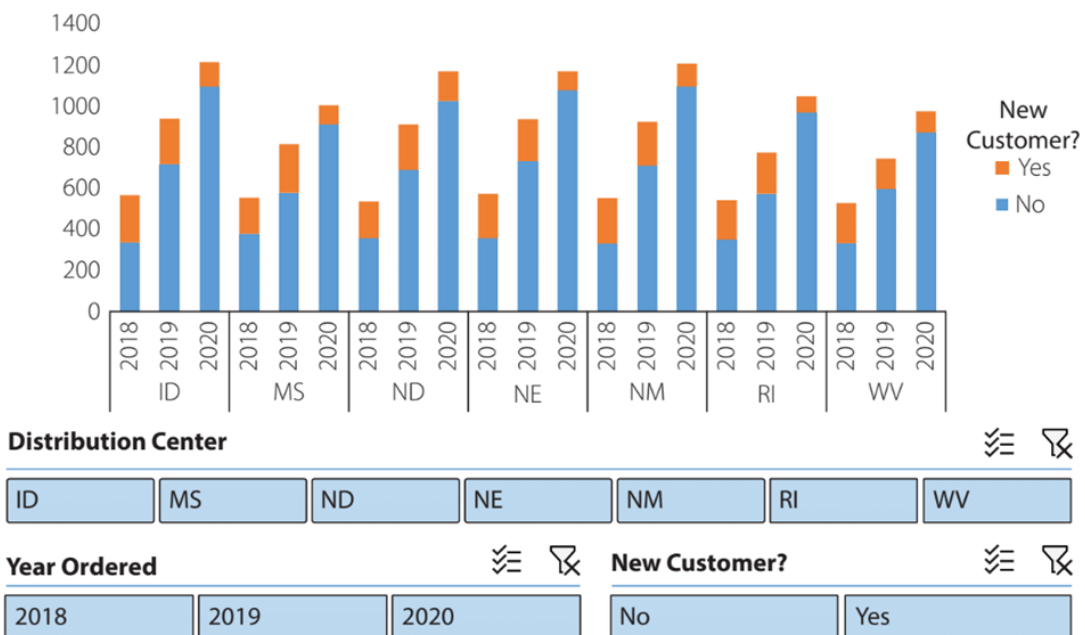
The borders can be removed from a Slicer box by executing the following steps to create a new slicer style which can be applied to any Slicer in the Excel workbook.

- Step 14.
 - With a Slicer box selected, click the Slicer tab on the Ribbon
- Step 15.
 - In the Slicer Styles group, right-click the current style and select Duplicate
- Step 16.

- When the Modify Slicer Style dialog box appears, select Whole Slicer from the Slicer Element: box and click Format.
- In the Format Slicer Element dialog box, click the Border tab and select None in the Presets area
- Click OK

Edit the chart to improve its appearance. The result should look similar to the following figure.

Total Sales (Thousand \$) by Distribution Center, Year Ordered, and New Customer



By selecting NM and RI in the Distribution Centre slicer, 2019 and 2020 in the Year Ordered slicer, and Yes in the New Customer? Slicer the following chart can be produced.

Total Sales (Thousand \$) by Distribution Center, Year Ordered, and New Customer



Using the slicers to explore the data answer the following questions:

- Which distribution centre showed the largest percent increase in total sales from 2018 to 2019 among new customers?
- Which distribution centre showed the largest percent decrease in total sales from 2019 to 2020 among new customers?
- Which distribution centre showed the largest percent increase in total sales from 2018 to 2020 among all customers?

Activity 2 – Association Between Two Quantitative Variables

A group of researchers followed for 40 years, a cohort of individuals born at a Copenhagen hospital between 1959 and 1961. The main conclusion of the study, published on an international peer-reviewed article of medicine, was that those individuals who drank red wine were richer, better educated, and had lower levels of cholesterol ratio (LHL/HDL), and consequently a lower risk of heart attacks than those who did not.

- Can we infer from this study that drinking more red wine can improve our cholesterol ratio, and therefore reduce the risk of heart attacks? Explain your answer.

- How could we improve the study if we wanted to investigate whether red wine consumption can lower cholesterol ratio, and therefore reduce the probability of heart attacks?

Activity 3 – Choosing an Appropriate Chart

A fitness lab has conducted an experiment in which 100 participants engage in a 30-minute period of high-intensity interval training (HIIT) and then are asked about their perceived level of exertion. They must select an answer from 1 to 4, where 1 = “did not feel challenged”; 2 = “broke a sweat but could have been pushed more”; 3 = “felt challenged but not overwhelmed”; and 4 = “extremely fatigued.” In addition, each participant’s body fat percentage is recorded. The following table displays a portion of the data from the experiment.

Participant	Body Fat Percentage	Exertion Level
1	27	3
2	31	4
3	24	1
4	21	2
5	14	4

Discuss what would be a good way to show how body fat percentage is related to exertion level?

Activity 4 – Correlation Analysis of Wide Data

Helen Wagner, a marketing analyst for Meredith Corporation, is examining a data set based on market research of potential patrons of a new social-media-based e-magazine. The data set has dozens of quantitative variables measuring characteristics such as patron income, age, daily hours spent on social media, money spent based on banner ads on websites, etc. Helen is interested in exploring the association between these variables.

Discuss possible visualisations and justify your recommendation.

Activity 5 – Types of Missing Data

Discuss each of the following descriptions based on the type of missing data being described: missing completely at random (MCAR), missing at random (MAR), or missing not at random (MNAR)

- a. A survey is conducted that asks a sample of people their age, highest level of education completed, and preference of political party. The results indicate that many entries for highest level of education completed are missing, and that respondents were more likely to omit this response if they had a high school education or less.
- b. An economic study is conducted asking people to report their profession, job title, and city of residence. Approximately 10% of the responses for job title are missing, and it is observed that men are less likely to provide their job title than women.

Activity 6 – More Univariate Analysis of EJB Data

In the EJB example discussed within the modules this week, univariate analysis was demonstrated on some of the variables. Following on, use a PivotChart to:

- a. construct the relative frequency distribution of records over the values of the Category variable. Describe your findings.
- b. construct the relative frequency distribution of records over the values of the New Customer variable. In the PivotTable, relabel a “No” value for New Customer as “Existing” and a “Yes” value as “New.” Describe your findings

Activity 7 – Association in Presence of a Categorical Variable

Consideration of a categorical variable can be added to a scatter chart displaying the relationship between two quantitative variables. Using the *NYC* data file, create a scatter chart for all 55 observation pairs of Median Monthly Rent and Percentage College Graduates. To add information about the borough to the observation pairs, the values of the Borough variable can be plotted as different data series.

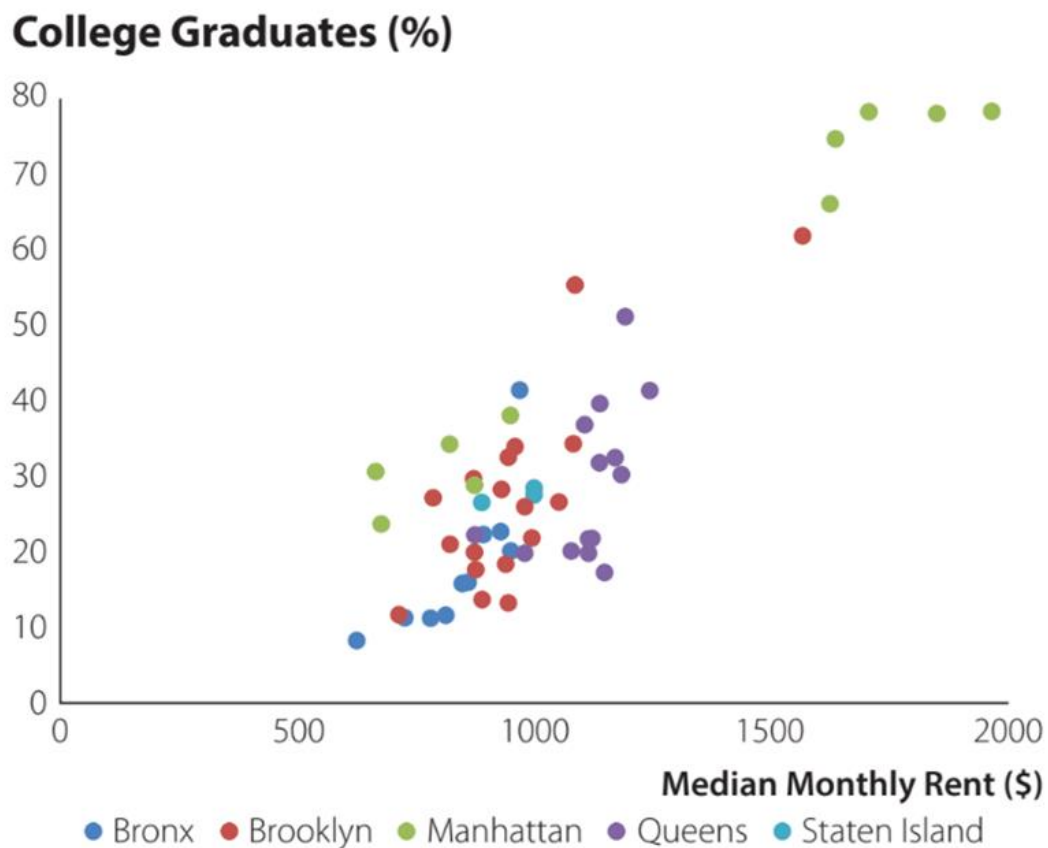
Open the *NYC* data file and complete the following steps:

- Step 1.

- Select cells A1:F56
- Step 2.
 - Click the Data tab on the Ribbon
- Step 3.
 - Click the Sort button in the Sort & Filter group
- Step 4.
 - When the Sort dialog box appears:
 - Select the check box for My data has headers In the Sort by row:
 - Select Borough for the Column entry
 - Select Cell Values for Sort on entry
 - Select A to Z for Order entry
 - Click OK
- Step 5.
 - Select cells B2:C11 (rows corresponding to the Bronx borough)
- Step 6.
 - Click the Insert tab on the Ribbon
- Step 7.
 - Click the Insert Scatter (X,Y) or Bubble Chart button in the Charts group
- Step 8.
 - When the list chart subtypes appear, click the Scatter button
- Step 9.
 - Right-click the chart and choose Select Data
- Step 10.
 - When the Select Data Source dialog box appears:
 - Select Series1 under Legend Entries (Series)
 - Click the Edit button
- Step 11.
 - When the Edit Series dialog box appears:
 - Enter Bronx in the box under Series name:
 - Click OK
- Step 12.
 - In the Select Data Source dialog box, click the Add button
- Step 13.
 - When the Edit Series dialog box appears
 - Enter Brooklyn in the box under Series name:
 - Enter =Data!\$B\$12:\$B\$29 in the box under Series X
 - Enter =Data!\$C\$12:\$C\$29 in the box under Series Y values:
 - Click OK
- Step 14.

- In the Select Data Source dialog box, click the Add button
- Step 15.
 - When the Edit Series dialog box appears
 - Enter Manhattan in the box under Series name:
 - Enter =Data!\$B\$30:\$B\$39 in the box under Series X
 - Enter =Data!\$C\$30:\$C\$39 in the box under Series Y values:
 - Click OK
- Step 16.
 - In the Select Data Source dialog box, click the Add button
- Step 17.
 - When the Edit Series dialog box appears
 - Enter Queens in the box under Series name:
 - Enter =Data!\$B\$40:\$B\$53 in the box under Series X
 - Enter =Data!\$C\$40:\$C\$53 in the box under Series Y values:
 - Click OK
- Step 18.
 - In the Select Data Source dialog box, click the Add button
- Step 19.
 - When the Edit Series dialog box appears
 - Enter Staten Island in the box under Series name:
 - Enter =Data!\$B\$54:\$B\$56 in the box under Series X
 - Enter =Data!\$C\$54:\$C\$56 in the box under Series Y values:
 - Click OK
 - Click OK to close the Select Data Source dialog box

Edit the chart to improve its appearance. The result should look similar to the following chart.



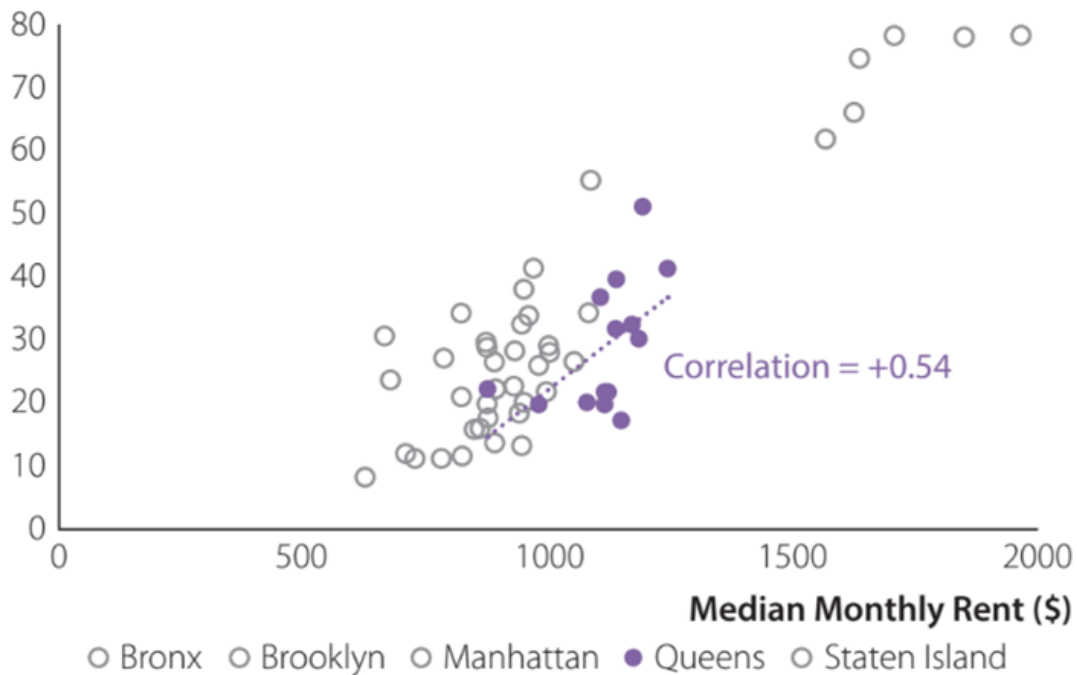
Activity 8 – Emphasising Relationships

Using the scatter chart you created in Activity 7, emphasize the relationship between Median Monthly Rent and Percentage College Graduates within the Queens by formatting all other boroughs with data points in grey with no fill. Create a trendline based only on the observations in Queens.

The result should look similar to the following chart.

Correlation between Rent and Percentage College Graduates in Queens Weaker than in Other Sub-boroughs

College Graduates (%)



Discuss what the trendline indicates about this relationship between Median Monthly Rent and Percentage College Graduates within the borough of Queens?

Activity 9 – Table Lens on Customer Data

Bravman Clothing sells high-end clothing products and is launching a service in which they use a same-day courier service to deliver purchases that customers have made by working with one of their personal stylists on the phone. In a pilot study, Bravman collected 25 observations consisting of the wait time the customer experienced during the order process, the customer's purchase amount, the customer's age, and the customer's credit score. The file *Bravman* contains the data for these 25 observations.

- Construct a table lens on this data sorted in decreasing order with respect to the values of the Purchase Amount variable.
- Summarize the relationships between Purchase Amount and the other three variables.

Activity 10 – Smartphone Sales

The file *Smartphone* contains data on the monthly sales revenue for a smartphone manufacturer.

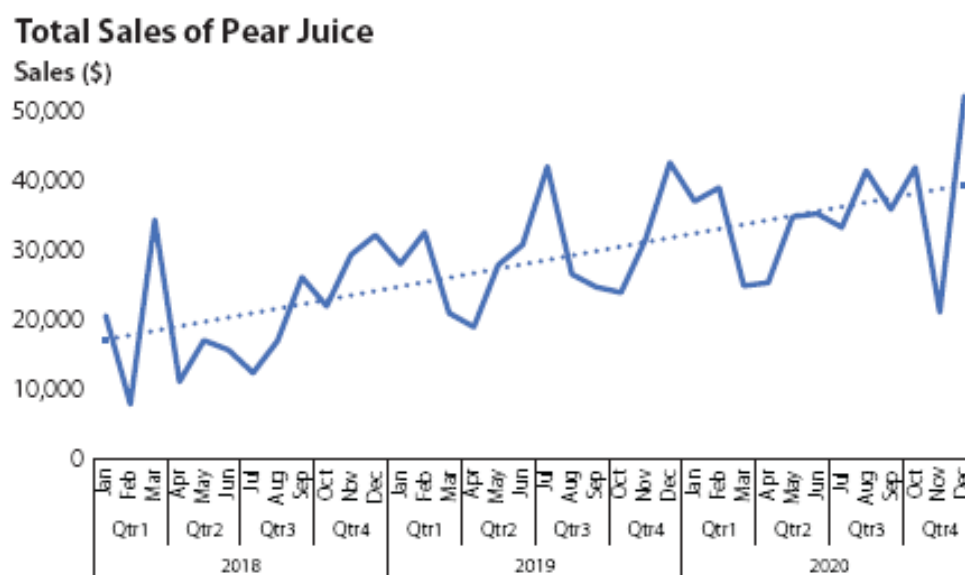
- Create a line chart to depict the sales time series at the annual level.
- Create a line chart to depict the sales time series at the quarterly level.
- Create a line chart to depict the sales time series at the monthly level.
- Discuss the insights that each of these three views provide.

Activity 11 – Creating a Moving Average Trendline

In time series data, it is often desirable to smooth the data in a manner that:

- only uses observations known up to the point of time at which the smoothing estimate is calculated, and
- does not use observations from the distant past.

Open the *PearJuiceEJB* data file, create a line chart with a linear trendline and edit the chart to improve its appearance. The result should look similar to the following chart.

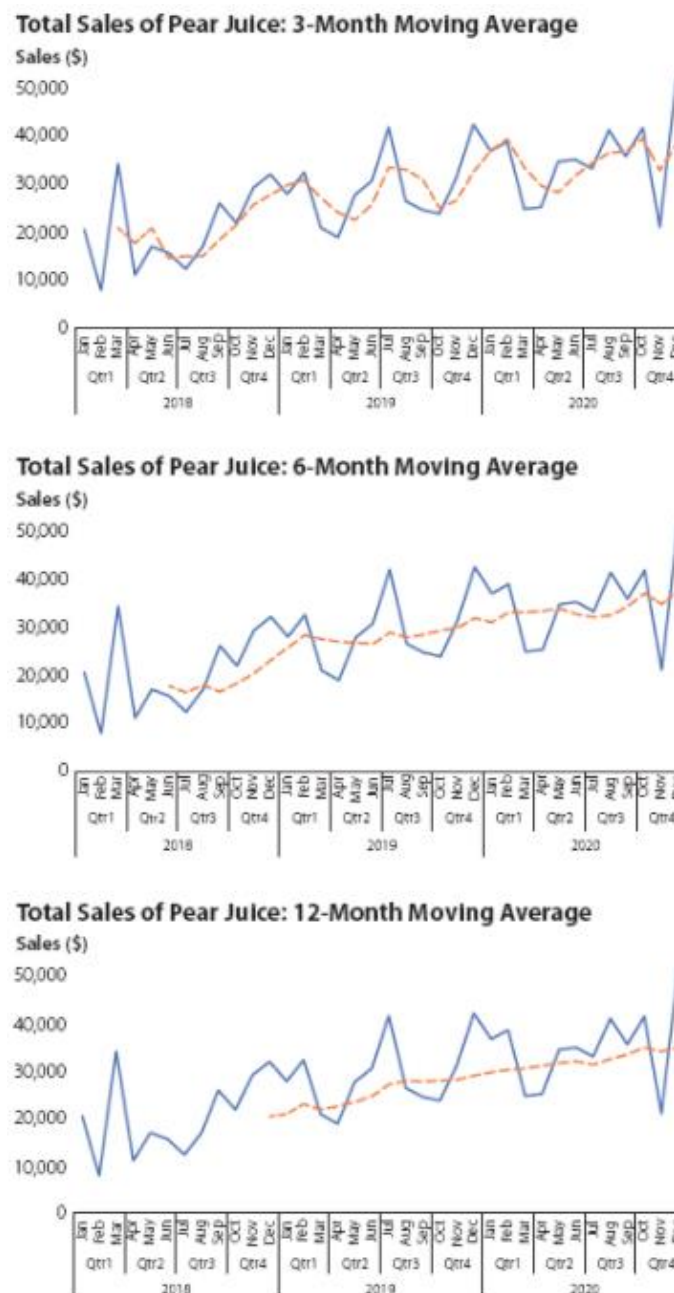


Now complete the following steps to add a three-month moving average trendline to the monthly pear juice chart.

- Step 1.
 - Select the data series by clicking on the data points.

- Step 2.
 - Right-click and select Add Trendline
 - In the Format Trendline task pane, under Trendline Options , select Moving Average and enter 3 in the Period box
- Step 3.
 - Repeat the preceding steps to create 6-month and 12-month moving average trendlines
- Step 4.
 - Create a trellis display of the Pear Juice Sales Smoothed by Moving Averages over 3 Months, 6 Months, and 12 Months

The result should look similar to the following trellis display.



Activity 12 – Umbrella Sales

The file *Umbrella* contains data on the quarterly sales revenue for a manufacturer of umbrellas and other weather-resistant gear.

- a. Create a line chart for the sales time series data. Add a four-period moving average to the chart to smooth the data.
- b. To investigate the possibility of seasonality on a quarterly basis, plot the data as a collection of five data series (one for each year). From this visualization, do you observe any indication of seasonality? If so, describe it.